

# Amit Kumar Mohanty

## AI / MACHINE LEARNING ENGINEER

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### PROFESSIONAL SUMMARY

AI/ML Engineer with **2 years of experience** taking machine learning systems from research into production — **LLMs, Agentic AI, computer vision**, and the data pipelines behind them. Named **"Rookie of the Year 2025"** at **Keywords Studios** for beating every performance target, and pursuing an **M.Tech in AI & ML at BITS Pilani**. Co-author of a peer-reviewed computer-vision paper. **Marathon runner** — the same patience and consistency I bring to systems that keep working long after launch.

### TECHNICAL SKILLS

**Languages:** Python, C++, SQL, Git, Data Structures & Algorithms, OOP

**ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, Deep Learning, Computer Vision, OpenCV, Transformers, Hugging Face, CNNs, Vision Transformers, BERT, NLP, Model Evaluation & Optimization

**LLMs & Generative AI:** Fine-tuning (LoRA, QLoRA), RAG, Agentic AI, LangChain, LangGraph, Prompt Engineering, Vector Databases (FAISS, Chroma), Gemini & OpenAI APIs

**MLOps & Cloud:** Docker, Kubernetes, CI/CD, MLflow, FastAPI, Model Serving & Monitoring, Data Pipelines, AWS, GCP (Vertex AI), Pandas, NumPy, MySQL, MongoDB

### EXPERIENCE

**Research Associate – AI, Keywords Studios India** — Gurgaon, Haryana

May 2025 – Present

- Train and improve **AI agents that finish multi-step jobs on their own** — booking flights, placing online orders, and pulling data from websites — making them far more reliable at completing tasks end to end.
- Helped build a **web-automation agent** that reads what is on the screen, takes a plain-English instruction, and then clicks, types, and navigates a browser to get the job done; tested and scored its behavior with a global team to **raise action accuracy** before every client release.
- Built custom data pipelines and pushed **85% of the training data I generated straight into production** — the first person on the global team to do this — which cut the time needed to get models ready.
- Named **"Rookie of the Year 2025"** for beating every performance target across model training, data pipelines, and cross-team work.

**Artificial Intelligence Intern, Infosys** — Remote

Nov 2024 – Feb 2025

- Built an **AI call-center assistant** with LangChain and LangSmith that answered customer questions **35% more accurately**, and cut response time by **25%**.
- Tested different prompt designs with automated LangSmith scoring and **reduced made-up answers (hallucinations) by 20%**, making conversations feel more natural and trustworthy.
- Worked with **12 engineers** on end-to-end ML workflows; selected to present at Infosys' **"Best Project by Indian Interns"** showcase.

**Data Analytics & ML Intern, QL2 Software** — Gurgaon, Haryana

Jun 2024 – Aug 2024

- Built **prediction models for competitive pricing** and automated the data collection that fed them, turning large raw datasets into clear business insights.

### PROJECTS

**Dynamic Screen Companion** — Python, FastAPI, Gemini API, OCR, TTS

Jun 2025 – Present

- Built a **real-time multi-modal assistant** that watches live screen activity through the Gemini API and guides the user with continuous voice feedback.
- Designed the architecture end to end and built a **FastAPI backend serving concurrent inference streams**; tuned the OCR pipeline to hold **95%+ text-extraction accuracy** and cut manual task time by **40%**.

**Diabetic Optiscan** — PyTorch, Vision Transformer, PyWavelets, APTOS

Jan 2025 – Jun 2025

- Built an AI tool that **checks retina photos for early signs of diabetic eye disease**, and co-authored a **peer-reviewed research paper** on how it was designed and the accuracy it gained.
- Fine-tuned a **Vision Transformer (ViT-B/16)** on the APTOS dataset and added a custom wavelet step that improved **detection precision by 50%** over earlier methods.

**Currency Detection App for the Visually Impaired** — TensorFlow, OpenCV, Mobile

Jul 2024 – Dec 2024

- Trained an image classifier on **200,000+ banknote images (96% accuracy)** and shipped it in a mobile app with voice output to help users avoid cash fraud; shaped the features through interviews with visually impaired users.

### EDUCATION

**Birla Institute of Technology and Science, Pilani (BITS Pilani)**

Jul 2026 – Jul 2028

M.Tech in Artificial Intelligence and Machine Learning (pursuing) — NLP, Deep Learning, Generative AI, LLMs, Deep RL

**K.R. Mangalam University**

Jul 2021 – Jul 2025

B.Tech in Computer Science and Engineering (AI/ML Specialization), Gurgaon — GPA: 8.5/10

### CERTIFICATIONS

AI/ML for Geodata Analysis — ISRO | Data Science Professional Training — Ducat India (2024–25)